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RESEARCH-ARTICLE

Accurate Bandwidth and Delay Prediction for 5G Cellular Networks

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The fifth-generation (5G) has empowerd various applications. Effective bandwidth and delay prediction in 5G cellular networks are essential for many applications, such as virtual reality and holographic video streaming. However, accurate bandwidth and delay prediction in 5G networks remains a challenging task due to the short-distance coverage and frequent handover properties of 5G base stations. In this paper, we propose HYPER, a hybrid bandwidth and delay prediction approach that uses an Auto Regressive Moving Average (ARMA) time series predictive model for intra-cell prediction and a Random Forest (RF) regression model for cross-cell prediction. Our ARMA model takes prior information as its input, while the RF model further uses related network and physical features to predict future performance. We conduct a measurement study in commercial 5G networks to analyze the relationship between these features and bandwidth/delay. Moreover, we also propose a handover window adaptation algorithm to automatically adjust the handover window size and determine which model to use during handover for accurate bandwidth and delay prediction. We use commercial 5G smartphones for data collection and conducted extensive experiments in diverse urban environments. Experimental results show that HYPER can reduce the prediction error by more than 13% compared to state-of-the-art prediction approaches.

CCS Concepts: • **Networks → Network performance modeling;**

Additional Key Words and Phrases: 5G cellular networks, bandwidth prediction, delay prediction, hybrid model

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1 Introduction

The **fifth-generation (5G)** has attracted extensive attention in the communication industry and academia in recent years. Compared to 4G LTE, a major improvement of 5G **New Radio (NR)** is its significantly higher bandwidth. In theory, 5G can deliver up to 20 Gbps peak downlink bitrates and more than 100 Mbps average data rates [26]. As a result, 5G enables a wide range of applications with high bandwidth and low delay requirements such as virtual reality, cloud gaming, and holographic video streaming. For these applications, being aware of the available bandwidth and delay in advance can help achieve adaptive configuration (e.g., rendering granularity, image resolution, etc.) [16] to improve the user experience.

Existing short-term performance prediction approaches in cellular networks can be divided into two categories. One is based on time series predictive models [24, 27, 41], another is based on **Machine Learning (ML)** regression models [17, 18, 34, 37, 39]. The former takes a time series of past information (e.g., past throughput) as the inputs to predict cellular link performance while the latter aims to further use related features, including upper-layer information (e.g., loss rate) and lower-layer information (e.g., signal strength, **Signal-to-Noise Ratio (SNR)**, link quality), to train a regression model. However, we find that both approaches cannot be directly applied in 5G cellular networks with **Commercial-Off-The-Shelf (COTS)** devices. ML regression models suffer from low-fidelity PHY/MAC layer information obtained using COTS 5G smartphones, leading to relatively worse prediction accuracy. Time series predictive models usually work well for relatively stable networks. However, link bandwidth in cellular networks is highly dynamic, especially for 5G networks. We have conducted a microscopic measurements study in a driving scenario (see details in Section 2). As shown in Figure 1, the short-range coverage of 5G base stations will introduce frequent handover events. A handover event may cause significant throughput and delay variations, making the time series model ineffective for 5G bandwidth prediction.

In this paper, we propose **HYPER**, a **hybrid bandwidth and delay prediction** approach for 5G cellular networks that combines time series predictive models and ML regression models. HYPER aims to achieve accurate bandwidth and delay prediction for 5G cellular networks with frequent handover events. Specifically, we first apply a light-weight **Auto Regressive Moving Average (ARMA)** time series predictive model to estimate the intra-cell bandwidth/delay time series with significant temporal correlations, taking prior bandwidth/delay usage of the current cell as the input. To further eliminate the prediction drift caused by significant throughput changes due to frequent handover, we use a **Random Forest (RF)** regression model to refine the bandwidth/delay prediction results during base station handover. However, accurate bandwidth prediction is still challenging from two aspects: (1) What features extracted from commercial devices can help improve bandwidth/delay prediction accuracy? Besides the commonly used network features, 5G link bandwidth/delay may be affected by a multitude of physical factors, such as moving speed and handover events [22]. (2) How to determine when to use the ARMA model and when to use the RF model during handover? We aim to use the ARMA model for bandwidth/delay prediction as soon as possible after a handover event due to its good property of capturing the temporal correlations of time series. However, there is a tradeoff in determining the size of handover windows[1] whose data are used for learning the ARMA parameters and simultaneously fed into the RF model for bandwidth prediction. A small window size may include insufficient training data for the ARMA parameters, while a larger one will lead to excessive use of the RF model. Both cases will lead to a reduction in the overall prediction accuracy.

[1]For each cell, its handover window is defined as the period whose bandwidth will be affected by the handover event.

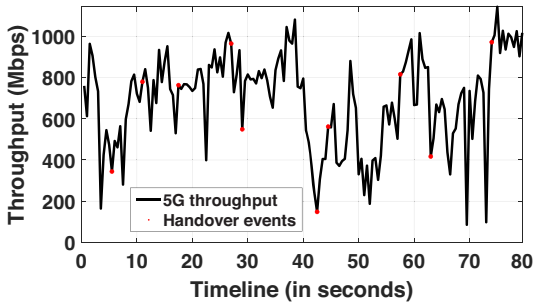


Fig. 1. 5G throughput and handover in a driving scenario.

We implement HYPER in COTS 5G smartphones without any change to the operating system or the need of rooting the phones. We mainly collect data in 5G cellular networks of **China Mobile Communications Group (CMCC)**, a major 5G carrier in China, and extract related information through APIs provided by the Android operating system. Our experiments consume more than one TB of cellular data in diverse urban areas with different movement states. For bandwidth prediction, experimental results show that HYPER can achieve a median downlink bandwidth prediction error of 10% in typical driving scenarios in urban areas, improving more than 13% compared to existing bandwidth prediction approaches. For the uplink, HYPER can achieve a median uplink bandwidth prediction error of 5%. For delay prediction, HYPER achieve a median prediction error of 21.1% in typical driving scenarios in urban areas, improving more than 3.8%-21.1% compared to existing approaches. We also integrate HYPER into a video streaming application to show that HYPER can benefit existing bandwidth-consuming applications. More importantly, overhead analysis results show that HYPER will not introduce much computation overhead during online bandwidth prediction.

In summary, we make the following key contributions:

(1) We have conducted an extensive measurement study in commercial 5G cellular networks to investigate the commercially available information that is related to link bandwidth and delay. We identify three upper-layer information, three lower-layer information, and two physical information that are significantly correlated with 5G cellular performance.
(2) We propose HYPER, the first hybrid bandwidth/delay prediction approach for 5G cellular networks. HYPER predicts the intra-cell bandwidth/delay using an ARMA model and the cross-cell bandwidth using an RF model. We propose a handover window adaptation algorithm based on time series' stationarity to adaptively determine the handover window size, which helps improve the prediction performance during handover.
(3) We evaluate HYPER using COTS smartphones in commercial 5G networks. Results show that HYPER achieves a better prediction accuracy than existing approaches in all the scenarios we have explored. The low system overhead of HYPER indicates that HYPER is lightweight for commercial 5G networks.

2 Background and Measurement Study

In this section, we first give the background of 5G infrastructure related to our experiments and then show some important observations that help the design of HYPER.

2.1 5G Infrastructure

5G uses multiple frequency bands including mmWave bands and microwave bands [28]. The mmWave bands at frequencies of 24 to 53 GHz provide a considerable amount of bandwidth,

which will bring extreme capacity, high throughput, and low latency. However, mmWave's short wavelength makes its signals vulnerable to attenuation and blockages [29], leading to significant data rate variation and shorter deployment distances between base stations. Microwave 5G radios operate at mid-band frequencies of sub-6 GHz whose radio signal largely remains omnidirectional and offers a decent data rate. In our experiments, we find that mid-band 5G radios are used by the CMCC carrier to form the basis of initial 5G services. However, to be compatible with higher frequency bands in the future, the distance between 5G base stations (≈200 meters) deployed by CMCC is still much shorter than that of existing LTE networks (≥500 meters) [1, 33], leading to frequent handover events.

Existing 5G infrastructure mainly includes two modes: **Non-StandAlone (NSA)** mode and **StandAlone (SA)** mode. NSA mode couples the 5G core network equipment with existing 4G LTE infrastructure and utilizes 5G NR for data plane operations while retaining the 4G infrastructure for control plane operations [22]. SA mode is fully independent of legacy cellular infrastructures. Its radio network and core network parts are both 5G NR. In our experiments, we adopt the SA mode that is supported by CMCC across the city.

In 5G networks, a base station is responsible for managing scheduling for both downlink and uplink channels for a **User Equipment (UE)**. In this paper, we focus on downlink channels as most data traffic in cellular networks is in the downlink direction [14]. Henceforth, link bandwidth in this paper refers to downlink bandwidth. While the investigation for the uplink channels is also important as the amount of user-generated content uploading data is increasing in the IoT area, we have also evaluated the uplink bandwidth prediction performance of HYPER in Section 5.3.

2.2 Measurement Study

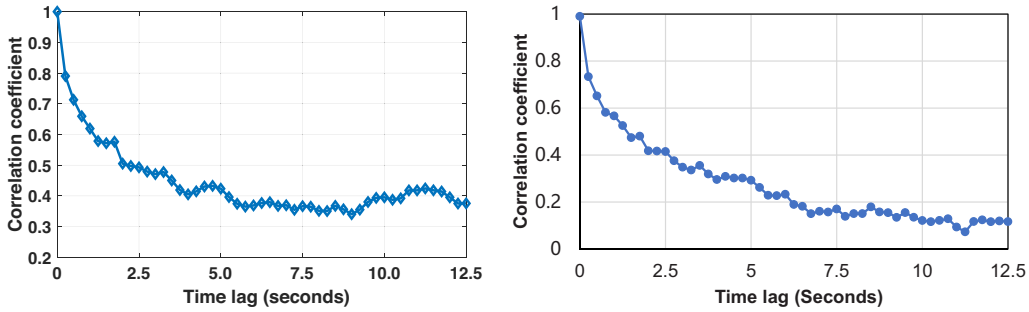
5G UE. We use two types of COTS 5G-capable smartphones, HUAWEI MATE30 5G (Hisilicon Kirin 990) and Xiaomi Mi10 5G (Qualcomm Snapdragon TM865), for 5G cellular data collection. Comparing their performance at the same locations, we have observed that the measurement results of these phones are consistent. We thus use MATE30 for all experiments. We confirm that despite 5G's high throughput, the device-side processing is not a bottleneck for MATE30, which is a high-end smartphone equipped with an eight-core CPU, 8 GB memory, Kirin 990 5G **System-on-Chip (SoC)** that fully integrates a Balong 5000 5G modem. In addition, MATE30 supports both 4G and 5G, allowing us to compare the throughput performance of these networks.

5G Monitoring Application. Android 10 [10] is the state-of-the-art Android OS and adds platform support for 5G NR related information. We are not aware of any commercial Android application that reports 5G-NR related information. Hence, we have developed a resource monitoring application based on Android 10 APIs for COTS 5G smartphones to obtain related information. We collect the following three categories of information at a sampling rate of 2 Hz to support our measurements: (1) Physical information includes system time, GPS, speed, and the connected cell ID. (2) Upper-layer information includes 5G service status, network interface and addresses, upload and download throughput,[2] packet loss rate,[3] RTT, and the variation of RTT. (3) Lower-layer information includes RSRP, RSRQ, and SINR.

Server and Experimental Scenario. We have investigated several cloud service providers in China and selected HUAWEI Cloud because it provides a maximum bandwidth of 2 Gbps, which is

[2]While the state-of-the-art Android OS does not support directly measuring throughput, we use the total traffic difference between adjacent time slots to calculate throughput.

[3]We seldom observe, however, losses of *ping* packets in our measurement traces. Almost all measurements in our traces have a packet loss rate of 0. Hence, we do not take the packet loss rate as a feature. As packet loss is a very informative and vital metric to reflect the current situation of the network, we can still easily use it as a feature of HYPER for bad 5G networks.



(a) Throughput. (b) Delay.

Fig. 2. Correlation coefficients among bandwidth/delay time series with different time lags.

much higher than the highest 5G speed we can obtain. We upload a 2 GB file to a server instance on HUAWEI Cloud. In our measurement study, we drive in the urban area at a speed of ~50 km/h and repeat the file download 15 times without interruption (i.e., 30 GB cellular data). In total, we recorded ~700 data for the measurement study.

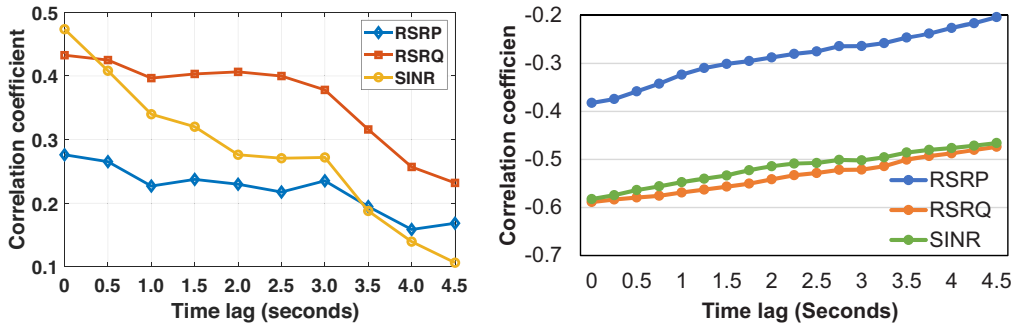
2.2.1 Observations. We have analyzed the correlations between link bandwidth/delay and a set of information.

Past information. Although link bandwidth in 5G networks is highly dynamic, past throughput and delay are still the most important information for predicting future bandwidth/delay. Figure 2(a) shows the correlation coefficients of link bandwidth time series with different time lags l, where $l \in [0, 25]$ seconds. Figure 2(b) shows the correlation coefficients of the delay time series with different time lags.

We can observe that the temporal correlation of link bandwidth time series and delay series are significant even at a lag of tens of seconds. Therefore, we use the ARMA model in HYPER since it is an effective tool to learn the function of previously observed values and random noise when the time series shows strong temporal correlations [4].

RSRP, RSRQ, and SINR. RSRP, RSRQ, and SINR are PHY/MAC layer information. RSRP is the linear average (in watts) of the downlink reference signals across the channel bandwidth. It measures the absolute power of the reference signal. RSRQ is defined as N×RSRP/RSSI, where N is the number of resource blocks across the downlink spectrum band, and **Received Signal Strength Indicator (RSSI)** is the total power that a UE observes across the entire download frequency band. RSRQ indicates the quality of the received reference signal. SINR is defined as the power of a certain signal of interest divided by the sum of the interference power (from all the other interfering signals) and the power of some background noise. It measures the quality of wireless connections. In COTS smartphones, RSRP, RSRQ, and SINR are of low fidelity and are reported as integers using Android APIs.

Let $\{b_i\}$ and $\{p_i\}$ denote the link bandwidth and a PHY/MAC layer information of the i-th second, respectively. Figure 3(a) shows the correlation coefficients between each PHY/MAC layer information time series $\{p_i\}$ and link bandwidth time series $\{b_{i+l}\}$ with different time lags $l \in [0, 5)$ seconds. We can see that the correlation between each PHY/MAC layer information and link bandwidth is significant but will rapidly decrease with the increasing number of time lags. As a result, these data will easily become out-of-date for bandwidth prediction. We have also found that unlike the 4G LTE networks [37], the correlation between RSRP and link bandwidth is relatively less significant to that between the other two PHY/MAC layer information and link bandwidth. Inspecting the trace, we observe that the trends of RSRP and link bandwidth are not consistent



(a) Throughput. (b) Delay.

Fig. 3. Correlation coefficients between RSRP, RSQP, SINR, and link bandwidth/delay.

at times. This may be because RSRP values will also be affected by other distinct factors of 5G networks such as the orientation of 5G base stations (deployed in the form of panels) [22].

We also investigate the correlation coefficient between RSRP, RSRQ, SINR, and delay. The results are shown in Figure 3(b). Similar to the observation with link bandwidth, the correlation between RSRP and delay is also relatively less significant compared with other PHY/MAC layer information. Compared with bandwidth, delay is less correlated with these PHY information. This is because latency is more susceptible to other factors along the path, such as network congestion.

Cell ID (i.e., Handover Events). Similar to LTE networks, a UE can only connect to one base station in 5G cellular networks. In a continuous trace, a handover event happens when the cell ID changes, which means a UE disconnects from one cell and connects to another cell. There will usually be abrupt throughput changes during handover since different base stations may have different bandwidth capacity and resource scheduling algorithms, and serve for different numbers of UEs. Taking bandwidth prediction as an example, we present the potential of ARMA. Figure 5 plots an example of bandwidth prediction results based on ARMA models in both 5G and 4G networks. We have the following important observations: (1) The ARMA model can achieve good prediction accuracy within a specific cell. Most bandwidth prediction error can be less than 10%, which shows the effectiveness of the ARMA model for intra-cell bandwidth prediction. (2) The prediction error will significantly increase during handover. This is because the fitting procedure of ARMA assumes a stationary time series [2], while handover events will usually lead to significant throughput changes. (3) Comparing to the LTE networks, handover events happen more frequently in 5G networks. The number of spikes of prediction errors in 5G networks is more than twice that in 4G networks. Moreover, the bandwidth prediction error of 5G networks is significantly larger since the bandwidth of 5G networks is an order of magnitude higher than that of 4G networks. As a result, inaccurately predicted bandwidth will have a greater impact on 5G networks.

RTT and Variation of RTT (for bandwidth prediction). RTT is a typical network layer information and is highly related to link bandwidth. There is often a conflict between throughput and packet RTT [20]. We integrate *ping* command into the monitoring application to measure the RTT to the server. Figure 4 shows the cross-correlation between link bandwidth and the measurements and variances of RTT. We can see that RTT and its variance have a significant negative cross-correlation with link bandwidth. Similarly, the correlation will also rapidly decrease as the time lag increases.

Speed. Due to the density of 5G base stations and the orientation of 5G signals, moving speed will have a great impact on bandwidth and delay. Figures 6 and 7 show example trace of throughput/delay and speeds in a driving scenario using measurements collected from commercial

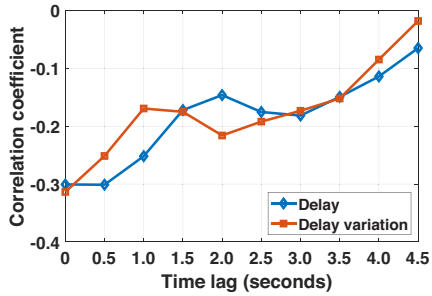
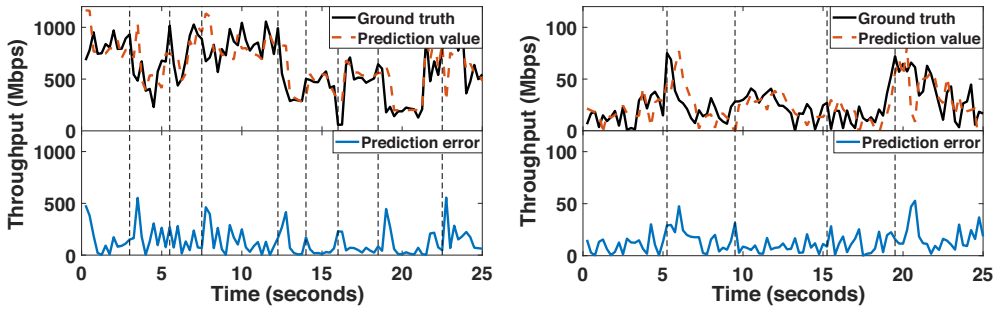


Fig. 4. Correlation coefficient between RTT, the variance of RTT, and link bandwidth.



(a) 5G networks (b) 4G networks

Fig. 5. An example of bandwidth prediction results using ARMA models in (a) 5G and (b) 4G networks. The upper figures show the predicted throughput and ground truth throughput. The bottom figures show the corresponding prediction error. Vertical dotted lines denote handover events.

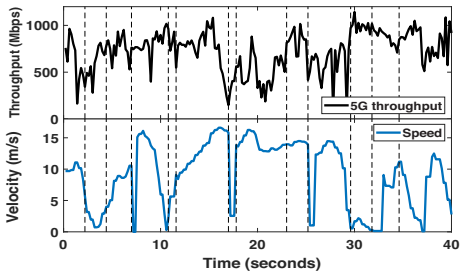


Fig. 6. 5G throughput and the corresponding moving speeds in a driving scenario. Dotted lines are the handover events.

5G networks. We also plot the handover events in the figure. We can see that: (1) Abrupt speed changes can easily cause significant bandwidth and delay changes. (2) When driving in urban areas, handover events can occur in less than ten seconds. As the average moving speed is around 10m/s, the distance between adjacent 5G base stations is usually less than 200 meters in our experiments. (3) Another interesting observation is that handover events usually occur when speed changes. This may be because the moving speed often changes at intersections in urban areas where there are usually more people and UEs sharing the spectrum, leading to more dynamic signal indicators (e.g., RSRQ, SINR) [33]. Besides, more base stations will be deployed at intersections and respond to handover events when their signal indicators are higher than that of the current serving cell [33].

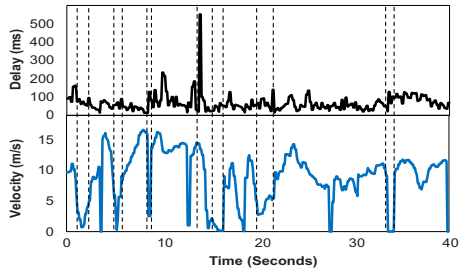


Fig. 7. Application delay and the corresponding moving speeds in a driving scenario.

Therefore, a handover decision will be more likely to be triggered at intersections with speed changes.

Summary. The above analysis shows that the intra-cell bandwidth time series has significant temporal correlations. However, time series predictive models alone cannot work well for 5G bandwidth/delay prediction due to the frequent handover events. On the other hand, we have identified several performance-related information that can be reported by COTS smartphones, including three upper-layer information (past throughput, RTT, the variation of RTT), three lower-layer information (RSRP, RSRQ, SINR), and two physical features (handover, speed). In HYPER, we aim to study how to use these features to help improve the bandwidth/delay prediction performance during base station handover.

3 HYPER Design

In this section, we first introduce the intra-cell ARMA model and cross-cell RF model, respectively. Then we will describe the hybrid prediction algorithm of HYPER. The core design of HYPER on bandwidth and delay prediction are the same. For brevity, we present HYPER with bandwidth prediction as an example.

3.1 Intra-cell Model

In HYPER, we use the ARMA model to learn the temporal correlation of intra-cell bandwidth time series. The general ARMA model can be modeled as ARMA(p, q), where p denotes p **AutoRegressive (AR)** terms and q denotes q **Moving Average (MA)** terms. Optimizing the performance of the ARMA prediction model requires tuning of the parameters p and q. For each base station, the data used to obtain its training parameters p and q is the bandwidth time series in the handover window. We tune the parameters p and q by a grid search while varying p and q between 0 and 3. We use the **Akaike Information Criterion (AIC)**, which considers both the goodness of fit and the simplicity of the ARMA model, for finding p and q [3]. The ARMA parameters p and q are retained for every step prediction within the same base station. For each step, we retrain the ARMA model and predict the bandwidth of the next time slot. This will not introduce much overhead since the ARMA model is lightweight and the throughput data for training is limited within a 5G base station.

3.2 Cross-cell Model

To compensate for the predicted values during handover, HYPER uses an RF regression model with related features for cross-cell bandwidth prediction.

3.2.1 Problem Formulation. The regression model takes the past related information as input to predict the bandwidth of the next time slot. In other words, the input to train the regression

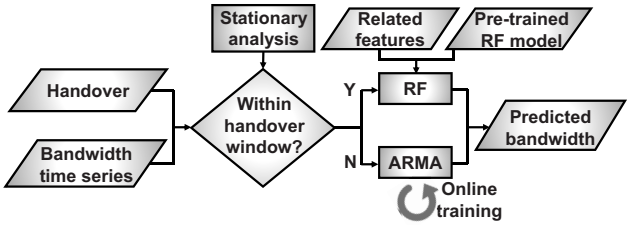


Fig. 8. Overview of HYPER and its workflow.

model is the feature vectors of the past W time slots. Each feature vector contains all the related features mentioned in Section 2.2.1 except for the cell ID information. Let F_t denote the feature vector at time slot t, an input vector I_t can be expressed as

$$I_t = [F_{t-1}, F_{t-2}, \ldots, F_{t-W}]. \tag{1}$$

The related features have different value ranges and are scaled down to the range of [0,1]. The corresponding train label is the bandwidth B_t at time slot t. Let $M()$ denote the prediction model learned offline. The regression problem can be formulated as

$$B_t = M(I_t). \tag{2}$$

3.2.2 Cross-cell Prediction. RF is a non-linear modeling technique used for time series regression. We select RF as our regression approach due to its capabilities of requiring low memory and computation overhead during online prediction, modeling complex relationships during handover, and avoiding overfitting.

In HYPER, we set $W = 1$ based on the experimental results, which means we use the input vector of the past single time slot to train the RF model. Detailed prediction performance using different numbers of past time slots will be shown in Section 5.6. Once the input vectors and the labels are constructed, we randomly select 70% of the total inputs as the training data and use the remaining 30% as the testing data. We perform a grid-based search on the training set to automatically find the optimal hyper-parameters of the RF model to maximize the classification accuracy. Once the RF model is constructed, we use the pre-trained model to predict the bandwidth during handover.

3.3 Hybrid Prediction Approach

To adaptively combine the above two models, we propose a hybrid prediction approach HYPER to achieve online bandwidth prediction. For a base station, HYPER first uses the pre-trained RF model to predict the bandwidth after the handover event. After determining the handover window size, HYPER will use the data in the handover window to learn the ARMA parameters p and q. Then HYPER will retrain the ARMA model at each time slot to predict the intra-cell bandwidth. Figure 8 shows an overview of HYPER.

It is non-trivial to determine the handover window size. A larger handover window size with more training data will lead to more accurate ARMA models but overuse the RF regression model. To adaptively determine the handover window size for different base stations, we propose a handover window adaptation algorithm based on the stationarity of bandwidth time series. Specifically, after a handover, HYPER will continuously check the stationarity of past bandwidth time series within the new base station. At the same time, HYPER will use the RF model to predict the bandwidth for these time slots. Once the time series becomes stationary, HYPER will determine the handover window size, learn appropriate ARMA parameters of the new base station, and switch

ALGORITHM 1: Hybrid Bandwidth Prediction Algorithm

Input: Time series of input vectors $\mathbf{I} = \{I_1, I_2, \ldots, I_t, \ldots\}$; Time series of bandwidth $\mathbf{B} = \{B_0, B_1, \ldots, B_{t-1}, \ldots\}$; Time series of connected cell ID $\mathbf{ID} = \{ID_0, ID_1, \ldots, ID_{t-1}, \ldots\}$, where ID_t is the connected cell ID at time slot t; Thresholds of ADF value, p-value and bandwidth changes TH_{ADF}, TH_p, TH_e; Minimum handover window size S_{min}; Pre-trained RF model $M()$

Output: Predicted bandwidth series $\hat{B} = \{\hat{B}_1, \hat{B}_2, \ldots, \hat{B}_t, \ldots\}$

```
startOfTrain ← 0, gotPAndQ ← False
for t = 1 to ∞ do
    if ID_t != ID_{t-1} or (B̂_{t-1} − B_{t-1})/B_{t-1} > TH_e then
        startOfTrain = t, gotPAndQ ← False, isStationary ← False
    if t − startOfTrain < S_min then
        B̂_t = M(I_t)
        continue
    if !isStationary then
        (V_ADF, V_p) = adfuller(B_[startOfTrain:(t−1)])
    if V_ADF > TH_ADF or V_p > TH_p then
        B̂_t = M(I_t)
    else
        isStationary ← True
        if !gotPAndQ then
            (p, q) = armaOrderSelect(B_[startOfTrain:(t−1)])
            gotPAndQ ← True
        armaModel = ARMATrain(B_[startOfTrain:(t−1)], p, q)
        B̂_t = armaModel(t)
```

to the ARMA model for subsequent bandwidth prediction. It is worth nothing that although the total time for handover procedures is small, e.g., less than 200ms [33], the throughput can be affected for several seconds after handover. This is because the communication resources may not be scheduled in time due to the complex cell load [35]. In our experiments, we also observe that the handover window size is usually in seconds.

HYPER uses a lightweight time series stationary analysis approach, **Augmented Dickey-Fuller (ADF)** test [7], to quantify the stationarity of a time series. The intuition behind the ADF test is that it can characterize how strongly a time series is stationary [9]. We can get two stationary metrics after running the ADF test on a time series: an ADF value V_{ADF} and a p-value V_p. For the ADF value, it should be a negative number. If the ADF value is less than a confidence value TH_{ADF}, we can assume the time series is stationary. On the other hand, a p-value below a threshold TH_p suggests the time series is stationary. In HYPER, we consider the bandwidth time series is stationary only when both metrics are less than the corresponding thresholds. In our current design, we carefully set TH_{ADF} to 1% confidence value and TH_p to 0.5 to ensure that HYPER captures a stationary time series with great confidence.

Algorithm 1 shows the details of our hybrid prediction algorithm. For each time slot, HYPER first uses the cell ID of the current serving cell to identify a handover event. Once a handover event is detected, HYPER needs to determine the handover window size of the new cell. It is worth noting that the intra-cell bandwidth may also change rapidly due to the nature of wireless communication and the radio resource scheduling [38]. Therefore, we consider that the relative bandwidth prediction error of the last time slot more than a predefined threshold TH_e will also trigger the handover window adaptation process. In HYPER, we empirically set TH_e to 1. Next,

we define a minimum handover window size S_{min} to eliminate unnecessary stationarity tests and ensure the amount of data for training the ARMA model. For each time slot after S_{min}, HYPER calculates the two stationary metrics V_{ADF} and V_p using the $adfuller$ function available in the Statsmodels package [25]. When the time series becomes stationary, HYPER sets the handover window size to the number of past time slots since handover. In practice, we observe that even in driving scenarios with fluctuating throughput, most bandwidth time series can be identified as stationary within three seconds after handover. As a result, we set S_{min} to 5 in our experiments. Detailed prediction performance of different S_{min} will be shown in Section 5.6. Then HYPER will use the bandwidth time series in the handover window to calculate the ARMA parameters p and q for the current cell. Finally, HYPER trains the ARMA model using these two parameters to estimate future bandwidth. Inside the handover windows, HYPER uses the pre-trained RF model to predict the bandwidth.

In HYPER, the RF model will not introduce much computation overhead. However, the ARMA parameter estimation process is relatively computation-intensive. On the other hand, we have also found that the ARMA parameters usually remain stable in a short time series. Considering both prediction accuracy and computation complexity, HYPER only estimates p and q once when the handover window size is determined for the new cell. Detailed prediction overhead will be shown in Section 5.8.

4 Theoretical Analysis

Taking delay prediction as an example, we try to provide deeper insights into the effectiveness of HYPER. Mathematically, the end-to-end delay can be presented as

$$T_k = T_k^{NR} + T_k^{DU} + T_k^{CU}, \tag{3}$$

where $T_k^{NR}, T_k^{DU}, T_k^{CU}$ is the delay k at the **New Radio (NR)**, **Distributed Unit (DU)**, and **Central Unit (CU)** of 5G [21], respectively. The delay between NR and DU, DU and CU areT_k^{DU}, T_k^{CU} are mainly influenced by the network condition and relatively smooth in a short period of time 5.7. The delay in NR is mainly influenced by the wireless link quality which is dynamic compared with other delay components.

Defined by the 5G specification [30], the 5G base station allocates transmission **resources block (RB)** and selects encoding **methods scheme (MCS)** according to **link quality indicator(LQI)** of the end user. When the end user selects MCS m in RB b, if the link quality exceeds the threshold of MCS m, it can obtain the corresponding data rate; otherwise, its data rate drops to zero. Mathematically, the achievable data rate can be modeled as:

$$r(t) = \begin{cases} r^m, & \text{if} \quad LQI \geq LQI^m \\ 0, & \text{otherwise} \end{cases}. \tag{4}$$

When the end user operates within a single cell (intra-cell), the achievable data rate changes only when the link quality exceeds or falls below a specific threshold. Higher link quality corresponds to a higher data rate in NR, thereby reducing the NR delay component T_k^{NR}. When the end user is intra-cell, its link quality rarely changes dramatically over a short period of time. The ARMA model of HYPER can make full use of the historical information to predict the delay. When the end user is cross-cell, its delay in NR is highly dependent on the PHY/MAC information, among other factors. The RF model of HYPER can capture their inherent connection and make an accurate prediction on delay. It is worth noting that the above explanation can also explain why HYPER is effective when predicting link bandwidth.

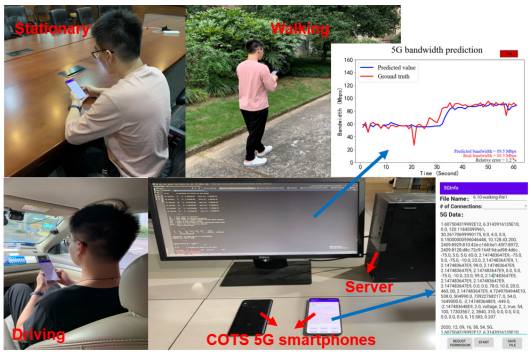


Fig. 9. Experiment scenarios, applications, and devices.

5 Evaluation

5.1 Implementation

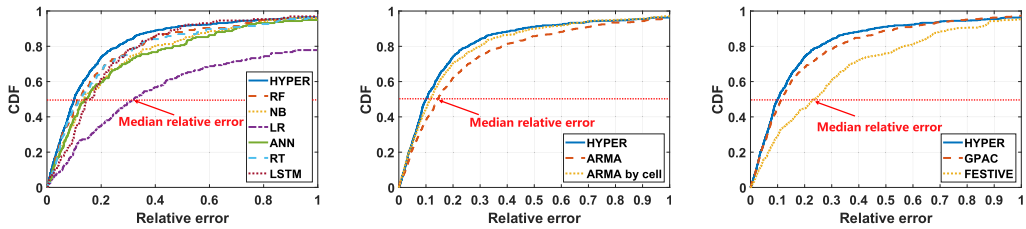
We implement the monitoring APP on commercial smartphones. From our measurement study, we have found that most features (e.g., handover, speed, and low-layer features) collected from commercial smartphones will only change every few seconds even in high-speed moving scenarios. Therefore, we limit the sampling rate in our monitoring application to second-level (i.e., 2Hz) and upload the collected information to a server every 500ms to remove redundant data and reduce computational overhead. We use a desktop with Intel i7-8700 CPU and 16GB of memory as the server to deploy HYPER. The server is responsible for processing the received information and returning the predicted bandwidth value to the UE. Then the UE can use the predicted bandwidth to make adaptive configuration for its applications. We provide a simple video streaming application in Section 5.2 to show how HYPER can help select an appropriate resolution to improve the user experience.

During each time slot (i.e., 500ms), the collected data size is around 150 bytes and the returned data size is around 8 bytes. In total, the data transmission overhead is around 2.6 Kbps and is negligible for commercial 5G networks. According to [33], the energy cost is around 1.3 mJ/s in HYPER and will not significantly deplete the smartphone's battery. Note that because HYPER uses lightweight prediction models, it can also run entirely on common smartphones at the cost of slightly extra energy consumption.

The 5G information of our work is mainly collected in urban driving scenarios during the daytime and night of 9 days starting from June 4, 2020. We also collect data in two other mobility scenarios: walking and stationary. For the walking scenario, the data were collected on a university campus ($0.7km \times 1.2km$). For the stationary scenario, we collected data in a meeting room ($8m \times 5m$) on the campus. All the three experiment scenarios and the data collection and analysis devices are depicted in Figure 9. Not surprisingly, we observe more handover events in driving scenarios than in walking and stationary scenarios. Detailed prediction performance in different mobility scenarios will be evaluated in Section 5.6. In summary, the collected datasets cover a wide range of scenarios, including different times of the day, locations, and movement speeds. The total amount of downloaded data is over one TB. The total numbers of unique cell IDs and handover events are 510 and 670, respectively. Table 1 shows the full dataset statistics. It is worth noting that in order to avoid model overfitting, we collect the dataset used for training and testing from various urban environments and diverse movement states and employ rigorous cross-validation techniques during model training to evaluate performance on unseen data. Furthermore, we carefully select the features (e.g., RSRP, RSRQ, SINR, RTT, and variation of RTT) based on extensive

Table 1. Details of the 5G Dataset

Data Samples	23,893 samples (per half-sec throughput with features)
Mobility and Duration	Driving (9 Days), Walking (5 Days), Stationary (2 Days)
Data Amount	1,130 GBs of 5G downloaded data
Cells and Handover	510 cells and 670 handover events



(a) Comparison with ML model-based approaches

(b) Comparison with ARMA model-based approaches

(c) Comparison with HTTP-based adaptive streaming protocols

Fig. 10. Cumulative Distribution Function (CDF) of bandwidth prediction relative error.

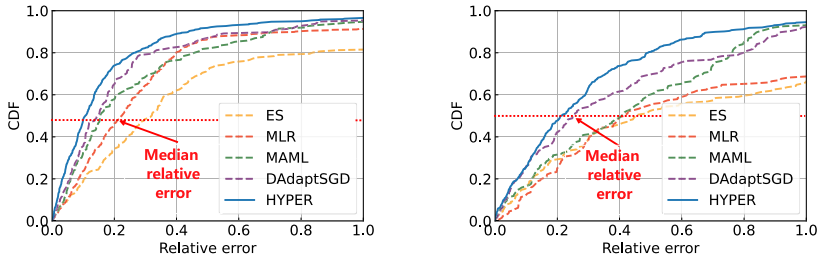
measurement studies that are widely applicable across different 5G environments to enhance the model's robustness and generalizability.

5.2 Prediction Results and Comparison Study

5.2.1 Comparison with ML Regression Model-based Approaches. We compare the prediction accuracy of HYPER with baseline prediction approaches that use **RF** [37], **NB**, **LR**, **ANN** [17, 18], and **RT** [34] ML regression models. It has been proved that training window size will not significantly affect the prediction accuracy [17, 37]. Considering that using a larger training window size will lead to a higher computation overhead, we set the window size to 1 in these approaches. We use all related features of the last time slot and current throughput to train these ML models. We carefully perform a grid-based search to automatically find the optimal hyper-parameters for constructing these ML models. We also implement a state-of-the-art deep learning model-based algorithm PERCEIVE [15] that uses **LSTM** for bandwidth prediction. We implement the LSTM neural network following the original paper and train the LSTM using the past 10s input of feature vectors.

Figure 10(a) shows that the median relative prediction error of HYPER is 10.0%, while the median relative prediction errors of ML regression models are between 11.5% and 30.2%. Compared to ML regression model-based approaches, HYPER improves the prediction accuracy by 13.0%-66.9%. These ML model-based approaches do not work well due to the coarse-grained features exposed from the Android OS. We also observe that the LSTM model does not perform well in our experiments. This is because the model needs millisecond-level information to achieve good prediction accuracy [15]. However, as mentioned earlier, most features collected from commercial smartphones will only change every few seconds and are second-level information. The LSTM model will easily include stale features due to the low sampling rate and frequent handover in commercial smartphones. In contrast, HYPER shows accurate bandwidth prediction results as it carefully considers the handover events and effectively learns the temporal correlations inside base stations.

5.2.2 Comparison with ARMA Model-based Approaches. We compare HYPER with the standard ARMA model [2]. The ARMA parameters p and q are also selected by a grid search while varying



(a) Bandwidth prediction performance. The median relative prediction error of HYPER is 10.5%

(b) Delay prediction performance. The median relative prediction error of HYPER is 21.3%

Fig. 11. Cumulative Distribution Function (CDF) of bandwidth prediction relative error.

the parameters for p and q between 0 and 3. For each time slot, we retrain the ARMA model using the throughput of the past five seconds. After significant throughput changes where the ARMA model cannot work, we will simply use the last throughput as the predicted value. Besides, we further implement an ARMA-by-cell model that only uses the past throughput of the current cell to train the ARMA models. In this model, we also use the last throughput as the predicted value during handover.

Figure 10(b) shows the prediction results of these two ARMA model-based approaches. As seen, the ARMA and ARMA-by-cell approaches achieve median relative prediction errors of 14.3% and 11.7%, respectively. Results show that using the past throughput in the same base station to predict the future bandwidth can achieve higher accuracy. Taking a closer look at Figure 10(a) and 10(b), the ARMA-by-cell model performs slightly better than the RF model, showing the effectiveness of the ARMA model in predicting 5G link bandwidth.

5.2.3 Comparison with Statistical Models and More Recent Machine Learning Approaches. In this section, we expand the comparison between HYPER and traditional statistical (i.e., **Exponential Smoothing (ES)** and **Multiple Linear Regression (MLR)**) and more recent machine learning approaches (i.e., Meta-learning (MAML) and DAdaptSGD [6]). The results are shown in Figure 11. When predicting the bandwidth, the median relative prediction error of HYPER is 10.5%, while the errors of ES, MLR, MAML, and DAdaptSGD are 32.04%, 22.45%, 15.56%, 14.74%, respectively. When predicting the delay, the median relative prediction error of HYPER is 21.3%, while the errors of ES, MLR, MAML, and DAdaptSGD are 45.79%, 41.78%, 40.85%, 25.30%, respectively.

There are two findings in the results: (1) Compared with traditional statistical models, more recent machine learning approaches (i.e., MAML and DAdaptSGD) perform better. This is because these approaches can capture complex, non-linear relationships in the data, which traditional statistical methods may struggle to identify. This capability enables more accurate predictions in scenarios where bandwidth and delay exhibit intricate patterns. (2) Compared with existing work, HYPER improves the prediction accuracy by 4.24%-24.49%. This is because HYPER carefully considers the handover events and effectively learns the temporal correlations inside base stations.

5.2.4 Comparison with HTTP-based Adaptive Streaming Protocols. Many popular HTTP-based adaptive streaming protocols will use historical bandwidth records to predict future bandwidth for video streaming [8, 11, 32]. We make a comparison study with two existing protocols. One is the GPAC's player [8] that uses the last seen bandwidth as the prediction value for the next

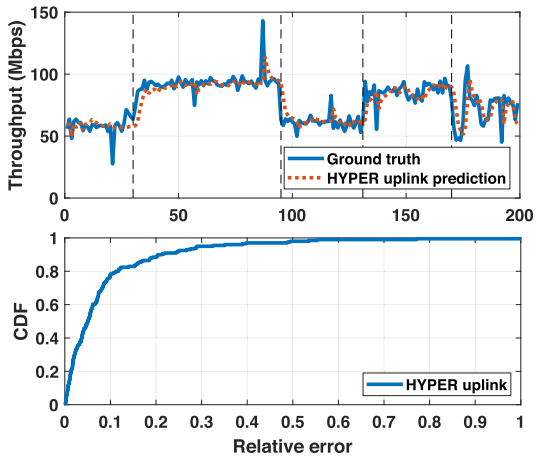


Fig. 12. Uplink bandwidth prediction performance of HYPER. Dotted lines are the handover events.

video segment. Another is FESTIVE [11], which uses smoothed harmonic mean of historical throughput measurements over a time window as the prediction. As recommended in [11], we use the harmonic mean over the last 20 samples to predict the future bandwidth.

Figure 10(c) shows the comparison results of HYPER and the two adaptive streaming protocols. Results show that compared to GPAC and FESTIVE, HYPER can improve the median relative prediction errors by 13.0% and 57.4%, respectively. In particular, the FESTIVE algorithm cannot perform well because it results in severe bandwidth under-utilization with the large bandwidth fluctuation of 5G networks.

5.3 Uplink Performance

We also collect a small uplink dataset with 1,000 samples to show the effectiveness of HYPER in 5G uplink. During the experiment, we continuously upload file data to the server and conduct walking tests on the campus with normal walking speeds. We use 80% samples to train the RF model and use the remaining samples to evaluate the uplink bandwidth prediction performance of HYPER. Figure 12(a) shows the ground truth and predicted uplink bandwidth. We can also observe that there are significant uplink throughput changes during handover. Figure 12(b) shows the corresponding uplink bandwidth prediction error. As seen, HYPER can achieve a median relative prediction error of 5.1%, which is significantly better than that (i.e., 10%) of 5G downlink. This is because, as shown in Figure 12(a), the bandwidth variation of uplink is much smaller than that of downlink, which makes the prediction results naturally more accurate.

5.4 Case Study 1: Video Streaming

We compare the prediction performance of HYPER, GPAC, and FESTIVE in a video streaming application. A UHD video with a 120 FPS frame rate is split into 0.5s segments and stored on the server. Each segment is encoded into four copies with different resolutions (1080P, 4K, 8K, 16K). For the four resolutions, we conservatively define the corresponding bandwidth ranges as [0,80], [80,320], [320,640], and [640,∞] in Mbps, to prevent video stalling due to remarkable bandwidth fluctuations. The client specifies the resolution to download each segment based on the predicted bandwidth. Figure 13 shows the percentages of downloaded segments with the best resolution when using different approaches for video streaming. Comparison results indicate that HYPER

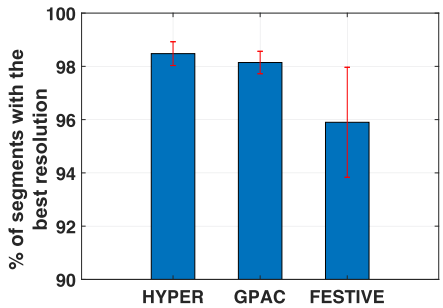


Fig. 13. Video streaming performance comparison. HYPER predicts the bandwidth accurately and has the most segments with best resolution.

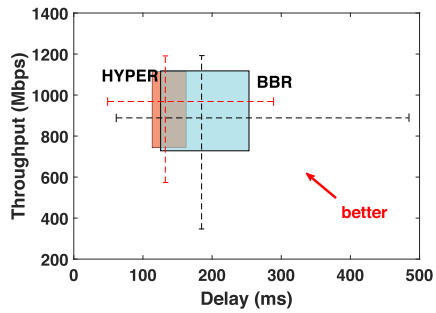


Fig. 14. Congestion control performance comparison. With predicted bandwidth, senders match their sending rate to achieve better performance.

can reduce the video stalling time by up to 2.6% for state-of-the-art adaptive video streaming applications using 5G networks.

5.5 Case Study 2: Congestion Control

Another important follow-up application of bandwidth prediction is congestion control. With an accurately predicted bandwidth, senders can match their sending rate to the available bandwidth capacity precisely and rapidly. Therefore, we propose an end-to-end HYPER-based congestion control algorithm that adaptively set the senders' rates to the predicted bandwidth of HYPER. We have implemented the HYPER-based congestion control algorithm to preliminarily evaluate the gains. We compare our algorithm with BBR [5], which is a leading congestion control algorithm.

Figure 14 shows the minimal, 25th, median, 75th, and maximal percentile throughputs and delays of our HYPER-based congestion control algorithm and BBR. We can observe that HYPER helps achieve lower delay while ensuring comparable throughput with BBR. This is because with an accurate predicted bandwidth information, our algorithm will introduce less packet congestion, thus reducing the average packet transmission delay. We can also observe that comparing to BBR, our algorithm achieves low variance in both delay and throughput by accurately estimating the bandwidth capacity.

5.6 Impact Factors

5.6.1 Impact of RF Window Size. The information in different numbers of past windows used to train the RF model will affect the prediction performance during handover. To evaluate the impact of RF window size, we use historical information in the past 1-5 time slots (i.e., 0.5s-2.5s) to train the RF model. Figure 15(a) shows the prediction error of different RF window sizes. Results show that adding more historical data does not improve the prediction accuracy. On the contrary, the accuracy may be degraded. The reason might be that historical related features can quickly become out-of-date in rapidly changing cellular networks. Using features further away in the past will degrade the prediction accuracy. Moreover, using a larger RF window size will lead to a higher computation overhead. Therefore, we set the RF window size to 1 in HYPER.

5.6.2 Impact of Minimum Handover Window Size. We evaluate the performance of different minimum handover window sizes S_{min} in Algorithm 1. An appropriate S_{min} can effectively reduce the prediction overhead and maintain a good accuracy of the ARMA model. We set the minimum handover window size to 3, 5, 7, 9, and 11 in our experiments. Figure 15(b) shows that the median relative prediction errors will increase when the window size is larger than 7. This is because

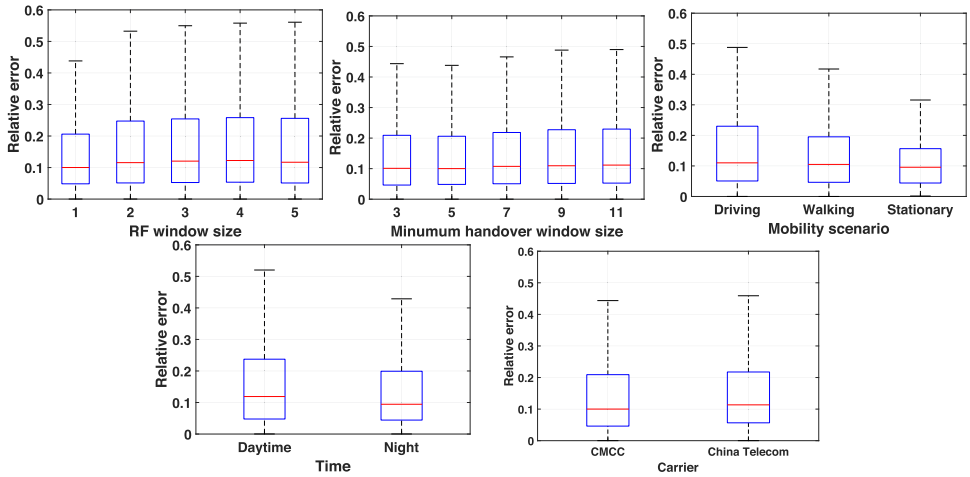


Fig. 15. Bandwidth prediction error of HYPER with different impact factors.

most of the final handover window size is within three seconds in our experiments. A large minimum window size will overuse the RF model in HYPER, thereby reducing the prediction accuracy. Considering that using a smaller minimum handover window size will introduce more computation overhead for stationary tests, we set $k = 5$ (i.e., 2.5s) in our experiments.

5.6.3 Impact of Mobility. As higher mobility scenarios will introduce more rapid link bandwidth changes and more frequent handover events, prediction accuracy may be greatly affected. To investigate the prediction performance in different mobility scenarios, we use the data collected in three different mobility scenarios, including a stationary scenario, a walking scenario, and a driving scenario. Interestingly, we observe that handover events occur even in the stationary scenario. This is because the signal indicator of the current serving cell may decrease due to many practical reasons, for example, some new UEs are accessing the cell. When the signal indicator of the current serving cell is lower than that of a neighboring cell, the handover procedure will be triggered. Figure 15(c) shows the bandwidth prediction errors in these scenarios. Results show that the overall prediction result under lower mobility scenarios (stationary and walking) is more accurate than that under driving scenarios. Interestingly, we can see that the median prediction errors of HYPER are similar in different mobility scenarios. This is because the 5G link bandwidth is highly dynamic even in stationary scenarios due to varying numbers of UEs in the base station, making it challenging to predict accurately.

5.6.4 Impact of Different Area. In order to evaluate the performance of HYPER more comprehensively, we extend the evaluation of HYPER in rural or suburban environments with different network characteristics. The results are shown in Figure 16.

Results show that the relative error of HYPER is 30.02% in suburban while in rural it is 16.98%. Compared with suburban and city, HYPER has more prediction accuracy in rural areas. There are mainly three reasons: (1) The density of 5G base stations is typically much lower in rural areas. As a result, users experience fewer handover events and the ARMA model can work more effectively because the time series is more stable and has fewer interruptions; (2) There are generally fewer obstructions (such as buildings, skyscrapers, or dense infrastructure) and fewer external sources of interference. This results in more consistent signal strength and better signal quality, leading to more stable throughput; (3) The user density is significantly lower. The available bandwidth per

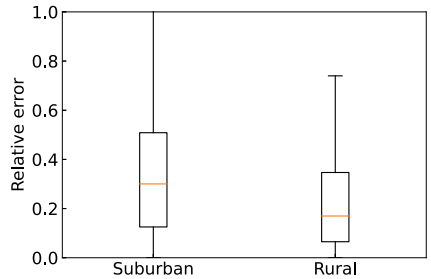


Fig. 16. Bandwidth prediction error of HYPER in suburban and rural areas. HYPER performs better in the rural area.

Table 2. HYPER Performance with Different User Density

User density	Low	Median	High
Relative error	10.71%	12.32%	17.14%

user is generally higher, and the fluctuations in bandwidth and delay due to network congestion or variable traffic loads are less severe, making it easier for HYPER to make accurate predictions.

5.6.5 Impact of Time. To evaluate HYPER's performance at different times, we use two datasets collected in similar driving scenarios during the daytime and night. Figure 15(d) shows that the median bandwidth prediction errors of daytime and night are 11.9% and 9.5%, respectively. Results show that HYPER performs better at night. This is because there are fewer cellular users at night and the bandwidth will be less affected by other UEs in the same base station.

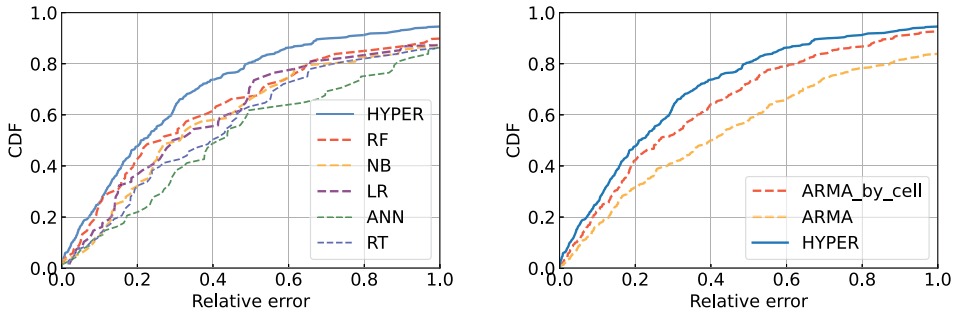
5.6.6 Impact of Carrier. We have also collected a small dataset (≈1,000 samples) in nearby 5G cellular networks deployed by China Telecom to investigate whether HYPER can be used for another carrier network. Figure 15(e) shows that HYPER can achieve comparable bandwidth prediction accuracy in different carrier networks. This is because the 5G networks deployed by CMCC and China Telecom in the city have similar NR network architecture and base station density.

5.6.7 Impact of User Density. In this section, we investigate the impact of highly dynamic network conditions on HYPER in terms of different user density. We evaluate the bandwidth prediction performance of HYPER in static scenario to avoid the impact of handover on the results. The results are shown in Table 2. HYPER performs better with a lower user-density. Specifically, in the scenario with low, median and high user density, the median relative error of HYPER is 10.71%, 12.32%, and 17.14%, respectively. This is because in low user density environments, there is typically less interference from other users. Besides, each user has higher available bandwidth. This allows for more stable bandwidth conditions, making it easier for the model to predict bandwidth accurately.

5.7 Performance on Delay Prediction

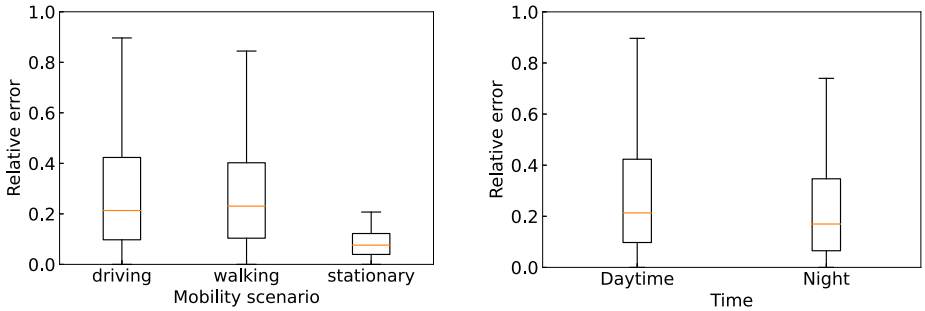
In this section, we evaluate the performance of HYPER on predicting delay. When not specified, the setup of the experiments are the same as that mentioned before.

Comparison with ML regression model-based and ARMA model-based approaches. We compare the prediction accuracy on delay of HYPER with different ML models (e.g., RF, NB, LR, ANN, and RT). Results are shown in Figure 17(a). The median relative prediction error of HYPER is 20.9%, while the errors of RF, NB, LR, ANN, and RT are 25.6%, 30.7%, 29.4%, 42.0%, and 30.3%, respectively. Figure 17(b) shows the prediction performance of two ARMA model-based



(a) Compared with RF, NB, LR, ANN, and RT. (b) Compared with ARMA and ARMA_by_cell.

Fig. 17. Cumulative Distribution Function (CDF) of HYPER's relative error.



(a) Different mobility scenario. (b) Different time.

Fig. 18. Delay prediction error of HYPER with mobility scenarios and time.

approaches. The median relative prediction of HYPER is 22.0% which is 12.9% and 3.8% lower than ARMA and ARMA-by-cell approaches.

Impact of mobility and time on prediction performance. In this part, we evaluate HYPER at different scenarios. Figure 18(a) shows the delay prediction errors with different speed. Compared with driving and walking scenario, the median delay prediction errors of HYPER in stationary scenario is much smaller which is only 7.6%. Figure 18(b) shows HYPER's performance at different times. HYPER performs better in daytime, whose median relative error is 3.2% lower than night.

There are two findings from the above experiments: (1) The observations for latency prediction are similar to those for bandwidth prediction. For example, HYPER performs better in stationary scenario and at night. These experiments further improve the credibility of our inference in Section 5.6; (2) Compared with bandwidth prediction, HYPER's prediction accuracy on delay is reduced. This is because delay is a sum of multiple components (e.g., T^{NR}, T^{DU}, and T^{CU}) while throughput is determined by the bottleneck of the entire link. Therefore, delay is influenced by more factors and varies more frequently compared with bandwidth. This observation is also consistent with the lower correlation coefficients (shown in Figure 2).

5.8 HYPER Overhead

As shown in Algorithm 1, the online prediction process can be divided into four phases: (1) Using the RF model within the minimum handover window during handover; (2) Running the ADF test and using the RF model outside the minimum handover window during handover where the time

Table 3. Prediction Delay of HYPER

Scenarios \ Phases		P1	P2	P3	P4	AVE
Driving	*Time*	7.035	23.021	1043.937	15.281	62.103
	Prop	27.6%	9.7%	4.7%	58.0%	
Walking	*Time*	7.654	9.525	921.842	10.179	37.093
	Prop	16.4%	3.3%	3.0%	77.3%	
Stationary	*Time*	6.969	8.173	722.629	1.321	9.674
	Prop	5.6%	1.5%	1.1%	91.8%	

Time is the prediction delay of each phase in milliseconds. *Prop* refers to the proportion of each phase in the total prediction trace.

series is not stationary; (3) Estimating the ARMA parameters, training the ARMA model, and predicting using the ARMA model at the time when the time series becomes stationary; and (4) Training the ARMA model, and predicting using the ARMA model for intra-cell time series. We insert timestamps in the code to estimate the prediction delay of each phase in different mobility scenarios. For each scenario, we run HYPER 100 times to calculate the average prediction delay.

Table 3 shows the proportion and prediction delay of each phase and the average prediction delays in three different mobility scenarios. As seen, the proportions of using the ARMA model and the RF model are different when using HYPER in different mobility scenarios. For low mobility scenarios, the ARMA model is preferred while for high mobility scenarios, the RF model will be used more. This is because higher moving speeds will lead to more frequent handover events, which will increase the frequency of using the RF model. Results also show that the prediction delays of most phases are within 30ms except for phase 3 due to the relatively computation-intensive ARMA parameter estimation process. In practical deployment, the delay of phase 3 can be significantly reduced if the ARMA parameters p and q of nearby base stations can be pre-measured. Comparing the average prediction delays of these three scenarios, the driving scenario is the worst because more time-consuming phases during handover (i.e., phase 2 and phase 3) will be included. However, the proportions of these time-consuming phases are less than 10% in all scenarios and will not introduce much prediction delay. We can see that the average prediction delays in all scenarios are less than 100ms, which is much smaller than the data collection interval in HYPER (i.e., 500ms). For video applications that usually prefetch video segments at segment-level (i.e., second-level) [32], bandwidth prediction can run at second-level. However, existing deep learning-based approaches [15, 23] require significantly more time to continuously retrain their models for diverse and changeable user behaviors. In contrast, HYPER is sufficient for online bandwidth prediction in existing bandwidth-consuming applications.

6 Discussion

6.1 Implications and Lessons Learned

In this section, we give a summary of the key observations, implications, and lessons learned from our experiments. Furthermore, we conduct a detailed error analysis to identify the main sources of prediction errors and present potential ways to further reduce these errors and improve the robustness of the model.

Observation 1: Existing bandwidth prediction approaches cannot work well in 5G networks while HYPER achieves a median bandwidth prediction error of around 10% on bandwidth prediction in all the scenarios we have explored.

Implication 1: Although the prediction accuracy of HYPER is better than state-of-the-art approaches, there is still a gap compared to the actual bandwidth and delay. Considering that cellular network performance may be affected by other commercially unobservable factors, such as radio resource scheduling and other UEs sharing the spectrum, the prediction accuracy can be largely improved with more detailed PHY-layer information exposed from the 5G chipset, such as the number of allocated resource blocks.

Observation 2: Handover can occur even in stationary scenarios.

Implication 2: This indicates that the ARMA model may not work even in stationary scenarios. It is a general approach to integrate ARMA and RF models to cooperatively predict the bandwidth for different scenarios.

Observation 3: The environment has a large impact on prediction performance. HYPER performs diversely at different times due to different numbers of UEs inside each base station and different traffic of each UE.

Implication 3: When predicting at different times or places, it is suggested to use features collected in similar scenarios to train RF models. To reduce the training cost, we can integrate transfer learning [36] into HYPER for cross-site training.

Observation 4: Stale throughput information of past base stations can affect the prediction accuracy of ARMA models.

Implication 4: Handover information should be used to train effective ARMA models for each cell.

6.2 Main Sources of Prediction Errors and Potential Improvements

Based on the experiments results, we summarize the causes of prediction errors as follows:

- Frequent handover events in 5G Networks: 5G networks have shorter cell coverage compared to 4G due to the higher frequency bands (e.g., millimeter-wave). This leads to frequent handover events, where **user equipment (UE)** switches between base stations. Handover events result in abrupt changes in throughput, which can significantly impact prediction accuracy.
- Dynamic nature of 5G and mobility: The bandwidth in 5G networks is highly dynamic, varying due to changes in user mobility (e.g., driving speed, distance from base stations), network congestion, and environmental factors (e.g., signal blockages in urban areas). The high speed of UEs (such as vehicles) can lead to sudden changes in signal quality.
- The lack of High-Fidelity PHY/MAC Layer Information on COTS Smartphones: On COTS smartphones, many PHY and MAC layer metrics are either unavailable or reported with low fidelity.

The potential ways to further reduce these errors and improve the robustness of the model are as follows:

- Mitigating the impact of frequent handover events: Improve the handover window adaptation algorithm by optimizing the transition between ARMA and RF models. One way to do this is by using reinforcement learning to dynamically adjust the handover window size based on real-time network conditions, learning from past prediction errors to improve future performance.
- Handling the dynamic nature of 5G bandwidth and mobility: A potential way of handling the dynamic nature of 5G bandwidth and mobility is incorporating more mobility-related information as a feature. Mobility data (e.g., speed, acceleration, direction) can be explicitly incorporated as an input feature for both time series and ML models. By using mobility-aware forecasting models, the system can adjust its predictions based on how fast the user

is moving and in which direction, allowing it to anticipate variations caused by velocity changes.
- Improving the fidelity of PHY/MAC layer information: Collaborate with network operators to access more accurate lower-layer information. By working with network providers, it might be possible to access better quality metrics through network-side APIs rather than relying on the limited data available from COTS devices.

7 Related Work

Bandwidth prediction in cellular networks has been extensively studied. We divide existing work into two major categories: (1) Time series predictive model-based prediction, and (2) ML model-based prediction. We then introduce some work on adaptive video streaming, which is a major use case that benefits from accurately predicted bandwidth.

7.1 Bandwidth and Delay Prediction for 5G/4G

Accurate bandwidth and delay is helpful for numerous mobile applications (e.g., AR, VR). However, compared to WiFi and wired networks, the links of 4G/5G cellular networks are highly dynamic, making it diffcult for accurate and real-time prediction. Nowadays, many reseachers exploit machine learning to achieve the goals [13, 17, 34, 37]. CapRadar [40] classifies bandwidth into scenarios and matches the optimal prediction model in the algorithm for each type of scenario. LinkForecast [37] identifies the most important features (from both upper and lower layers) and uses these features to predict link bandwidth in real time. PERCEIVE [15] exploits a 2-stage **LSTM (Long Short Term Memory)** design and makes throughput predictions for the next 100ms. Xu et al. [34] use **Regression Trees (RTs)** to predict the short-term network performance with correlated upper-layer information. Yue et al. [37] employ an RF model to train six upper-layer and lower-layer information for bandwidth prediction. Liu and Cerpa [17] propose three offline prediction methods (i.e., **Naive Bayes (NB)**, **Logistic Regression (LR)**, and **Artificial Neural Networks (ANN)**) to conduct link prediction with PHY/MAC layer information. Based on this work, they further propose an online LR model-based link estimator TALENT in [18].

However, we observe that lower-layer features collected from COTS smartphones **(Reference Signal Received Power (RSRP)**, **Reference Signal Received Quality (RSRQ)**, and **Signal-to-Interference Noise Ratio (SINR)**) are usually of low fidelity. To make it worse, some commonly used lower-layer information is even not available in smartphones. For example, although we can access **Channel Quality Indicator (CQI)** through an Android API, its reported value is always a maximum value of *int* type, which means CQI is still a developing function. Using ML models alone with these low-fidelity features will also lead to poor prediction performance.

Summary: In HYPER, we prefer to use time series predictive models for bandwidth prediction due to the higher prediction accuracy. However, while time series predictive models cannot deal with handover events, we need to use ML models to assist in predicting. HYPER can adaptively allocate these two types of prediction models for bandwidth prediction within appropriate periods.

7.2 Time Series Predictive Model-based Prediction

In cellular networks, time-stamped throughput data can be stored as time series and frequently show temporal correlations [4]. Historical throughput time series can be used for bandwidth prediction and network planning. Autoregressive models are common time series predictive models that have been extensively used in wireless networks. Papagiannaki et al. [24] use a low-order **AutoRegressive Integrated Moving Average (ARIMA)** model to capture the short

memory process of traffic loads, while Sadek and Khotanzad [27] use a Gegenbauer ARMA model to specify long memory processes. There are also some studies using similar models to predict other related metrics in cellular networks. For example, in [12] and [31], autoregressive models are applied to predict future SNR values and handover events, respectively.

However, these autoregressive models can only work well in relatively stable networks without abrupt throughput changes. In other words, ARMA models work well inside base stations but have a poor prediction performance during handover. Thus, these approaches cannot be directly applied in 5G cellular networks due to frequent handover events between short-coverage 5G base stations.

7.3 Adaptive Video Streaming

Adaptive video streaming helps reduce the video stalling time and improve user-perceived **Quality of Experience (QoE)**. Quality of adaptive streaming depends highly on the accuracy of clients' prediction of end-to-end network bandwidth. In LTE networks, extensive studies [8, 11, 19, 32, 42] have developed tools to predict available bandwidth and strive for a balance between high video quality (bitrate) and the risk of buffer underrun. For example, Xie et al. [32] propose an adaptive video streaming framework PiStream that uses LTE's PHY layer information to enable bandwidth-level prediction in LTE networks and agile video rate adaptation.

In this paper, we also compare HYPER with two bandwidth prediction approaches that have been integrated into existing HTTP-based adaptive video streaming protocols to show its possibility of helping improve the video QoE in 5G networks.

8 Conclusion

In this paper, we first conducted a primary measurement study in commercial 5G networks to learn the temporal correlations of bandwidth/delay time series and identify bandwidth-related features. We then propose HYPER, a hybrid bandwidth prediction and delay approach that combines an ARMA model and an RF regression model to predict intra-cell and cross-cell bandwidth and delay, respectively. We propose a handover window adaptation algorithm in HYPER to enable accurate prediction in 5G networks with frequent handover events. Extensive experiments using COTS smartphones in commercial 5G networks show that HYPER outperforms state-of-the-art prediction approaches. Results also show that HYPER can work well in various mobility scenarios without introducing much computation overhead.

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